Resilience Academy

SDI Training for ICT Professionals

Understanding Spatial Data & SDI

What is spatial data and why does Tanzania need a Spatial Data Infrastructure?

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Day 1 | 1.5 Hours | crd.resilienceacademy.ac.tz

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Resilience Academy

Session Overview

What we will cover:

1. What is spatial data? Real examples from Tanzania
2. Components every spatial dataset has
3. Two families: **vector** and **raster**
4. Common file formats you will encounter
5. Coordinate systems — just the essentials
6. What is an SDI and why it matters
7. The five pillars of any SDI
8. OGC standards in plain language
9. Tanzania's CRD platform: a real SDI
10. Live demo: browsing CRD

By the end you will...

- Explain spatial data to a colleague
- Recognise vector vs. raster data
- Know what an SDI does and why CRD exists
- Browse datasets on CRD confidently

What Is Spatial Data?

Simple Definition

Spatial data is any data that has a **location** attached to it — it answers the question *"Where?"*

Think of it this way:

- A spreadsheet with student names is *regular* data.
- The *same* spreadsheet with GPS coordinates of each school becomes **spatial data**.

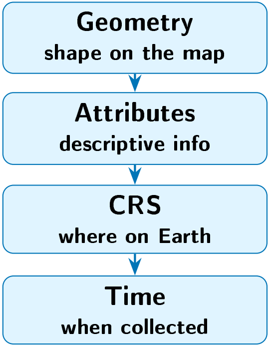
Everyday Tanzanian examples:

- Climate: Rainfall measurements across Dar es Salaam wards
- Business: Locations of registered shops in Zanzibar
- Infrastructure: Road network connecting Dodoma to regional towns
- Disaster risk: Flood zones mapped along the Msimbazi River
- Health: Locations of all health facilities in Tanga Region

Any time you see a map, the information behind it is spatial data.

Four Components Every Spatial Dataset Has

1. **Geometry** — the shape on the map
   A point, a line, or an area (polygon).
2. **Attributes** — the descriptive information
   Name, population, road type, flood depth...
   Think of it as the spreadsheet columns.
3. **Coordinate Reference System (CRS)**
   The "language" that tells software where exactly on Earth the geometry sits.
4. **Time** (optional but common)
   When was this data collected? Is it 2020 census data or 2025 satellite imagery?



Vector Data — Drawing on the Map

Vector data represents real-world features as **points**, **lines**, and **polygons**.

Think of it like drawing on top of a map with a pen.

Points

A single location.

Examples:

Schools, hospitals, business locations, weather stations

Lines

Connected sequence of points.

Examples:

Roads, rivers, power lines, bus routes

Polygons

A closed area.

Examples:

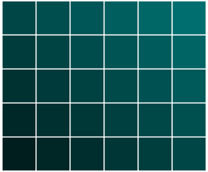
Wards, districts, flood zones, building footprints

Key idea: each feature carries an **attribute table** — just like a row in a spreadsheet.

Raster Data — A Grid of Pixels

Raster data divides the world into a **grid of cells** (pixels), each storing a value.

Think of a digital photo: millions of tiny squares, each with a colour.



Each cell = one value (e.g. elevation)

Common raster datasets:

- **Satellite imagery** — e.g. the `Tanga_imagery_reduced` dataset you will use later (a photo of the Earth from space)
- **Digital Elevation Models (DEMs)** — height of the land surface; used for flood modelling
- **Land cover maps** — each pixel classified as forest, water, urban, farmland…
- **Rainfall grids** — estimated rainfall per pixel from weather models

Common Spatial Data Formats

You do not need to memorise these — just recognise them when you see them.

Format	Type	Think of it as...
GeoJSON (`.geojson`)	Vector	A text file a browser can read
Shapefile (`.shp + friends`)	Vector	The old standard — needs 4–6 files together
GeoPackage (`.gpkg`)	Both	A modern single-file replacement
GeoTIFF (`.tif`)	Raster	A photograph with coordinates baked in
CSV with XY (`.csv`)	Tabular	A spreadsheet with latitude/longitude columns
KML/KMZ (`.kml`)	Both	Google Earth files

Practical Tip

On CRD you will mostly download **Shapefiles** and **GeoTIFFs**. QGIS can open all of the formats above — just drag and drop the file into the QGIS window.

Coordinate Systems — Just the Essentials

What is a Coordinate Reference System (CRS)?

A CRS is the “address system” that tells your computer *exactly* where something is on the round Earth displayed on a flat screen.

Two you need to know:

1. **WGS 84** (EPSG:4326)
   - Uses **latitude / longitude** (degrees)
   - What your **GPS phone** gives you
   - Good for *storing and sharing* data worldwide
2. **UTM Zone 37S** (EPSG:32737)
   - Uses **metres** as the unit
   - Designed for **Tanzania’s location**
   - Good for *measuring distances and areas*

Rule of Thumb

- **Sharing data online?**
  Use WGS 84
- **Measuring something?**
  Switch to UTM 37S

Don’t Panic

QGIS can convert between systems with two clicks. You just need to know *which* system your data is in.

What Is a Spatial Data Infrastructure (SDI)?

The Highway Analogy

Imagine spatial data as **goods** that need to move between cities (organisations).
An **SDI** is the **highway network** — the roads, signs, fuel stations, and traffic rules — that makes sure the goods arrive safely and everyone can use the same roads.

Without an SDI:

- Every organisation keeps data on its own hard drive
- No one knows what data already exists
- Duplicate effort, incompatible formats, outdated information

With an SDI:

- Data is published in one place with a **standard format**
- Anyone can **search**, **view**, and **download** what they need
- Organisations **collaborate** instead of working in silos

The Five Pillars of an SDI

Improved Decision-Making for Tanzania

Data	Metadata	Services	Standards	People & Policy
Maps, satellite images, survey data	Descriptions so you can *find* data	Web tools to view, download, search	Agreed rules so data works together	Organisations, laws, collaboration

Spatial Data Infrastructure

Remove any one pillar and the building is unstable — all five are needed.

OGC Standards in Plain Language

The **Open Geospatial Consortium (OGC)** created standard "languages" so different software can share maps. Here are the four you will encounter:

WMS — Web Map Service

Analogy: Asking for a *photo* of a map.
You get a picture (PNG/JPEG). You can look at it but cannot edit the data behind it.

WCS — Web Coverage Service

Analogy: Asking for a *raw satellite image*.
Like WFS but for raster data — you download the actual pixel values (e.g. elevation).

WFS — Web Feature Service

Analogy: Asking for the *raw data file*.
You get the actual vector shapes + attributes.
You can analyse, filter, and edit.

CSW — Catalogue Service for the Web

Analogy: A *search engine* for spatial data.
You search "flood maps Dar es Salaam" and the catalogue returns matching datasets.

You will not type these URLs yourself. CRD and QGIS use them behind the scenes.

CRD — Tanzania's Community Resilience SDI

What is CRD?

The **Climate Risk Database** is an online platform built on GeoNode that serves as Tanzania's SDI for climate and resilience data.

Key facts:

- **287+ datasets** covering flood risk, infrastructure, land use, climate, health, and more
- Hosted at **crd.resilienceacademy.ac.tz**
- Built on **GeoNode** (open-source SDI platform)
- Supports WMS, WFS, WCS, and CSW
- Data contributed by **multiple partners**: universities, government agencies, NGOs

What You Can Do on CRD

1. **Search** for datasets by keyword, region, or category
2. **Preview** maps in the browser
3. **Download** data in multiple formats
4. **View metadata** — who made it, when, what CRS
5. **Create maps** by combining layers online

This afternoon you will upload your *own* data to a GeoNode instance running on

Live Demo — Exploring CRD

Follow Along in Your Browser

Open **crd.resilienceacademy.ac.tz** and try these tasks:

1. **Browse the catalogue:** Click *"Datasets"* in the top menu. How many datasets are listed?
2. **Search by keyword:** Type **"flood"** in the search bar. What do you find?
3. **Preview a map:** Click on any flood-related dataset. Does the map load in the browser?
4. **Check the metadata:** On the same dataset page, scroll down. Who created the data? What CRS is it in? When was it last updated?
5. **Explore download options:** Look for the download button. What formats are available — Shapefile? GeoJSON? GeoTIFF?
6. **Filter by region:** Can you find datasets specifically about **Dar es Salaam** or **Zanzibar**?

Take 5 minutes. Write down one dataset you found interesting — we will discuss as a group.

Session 1 Summary

What we covered:

- Spatial data = data with a **location**
- Every spatial dataset has **geometry**, **attributes**, a **CRS**, and often a **time**
- **Vector** data: points, lines, polygons
- **Raster** data: grids of pixels
- Common formats: Shapefile, GeoJSON, GeoTIFF, GeoPackage
- Two CRS to remember: **WGS 84** (sharing) and **UTM 37S** (measuring)

SDI essentials:

- An SDI is the highway network for spatial data
- Five pillars: data, metadata, services, standards, people
- OGC standards let different tools talk to each other
- CRD is Tanzania's resilience SDI with 287+ datasets

Next Session

Session 2: Technology Overview

We look at the tools behind the scenes —